Ch 8.1: Decision Trees

Lecture 21 - CMSE 381

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Mon, Mar 28, 2022

Announcements UPDATE

Last time:

- Lab - Non-linearity

This lecture:

- 8.1 Decision Trees

Announcements:

- Mon 3/28: 8.1
- Weds 3/30: 8.2
- Fri 4/1: Lab: 8.3
  - HW #8 Due Friday

Section 1

Last time

The idea with polynomial functions

Learn the model:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_K x_i^K + \varepsilon_i$$

My data:

$$\overset{X}{\begin{pmatrix} 1 \\ 3 \\ 0 \\ 2 \\ 5 \\ 1 \\ 8 \end{pmatrix}} \qquad \overset{Y}{\begin{pmatrix} \quad \\ \quad \\ \quad \\ \quad \\ \quad \\ \quad \\ \quad \end{pmatrix}}$$

What I pass to `LinearRegression` or `LogisticRegression` as features:

$$\begin{array}{cccc} 1 & \mathtt{x} & \mathtt{x}^2 & \mathtt{x}^3 \\ \end{array}$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 27 \\ 1 & 0 & 0 & 0 \\ 1 & 2 & 4 & 8 \\ 1 & 5 & 25 & 125 \\ 1 & 1 & 1 & 1 \\ 1 & 8 & 64 & \end{pmatrix}$$

The idea with step functions

Knots $z_1 \leq z_2 \leq \cdots \leq z_K$. Let $c_j(x) = \mathrm{I}(z_j \leq x \leq \cancel{x}_{j+1})$. Learn the model:

$$y_i = \beta_0 + \beta_1 c_1(x_i) + \beta_2 c_2(x_i) + \cdots + \beta_K c_K(x_i) + \varepsilon_i$$

My data:

$$\begin{pmatrix} 1 \\ 3 \\ 0 \\ 2 \\ 5 \\ 1 \\ 8 \end{pmatrix}$$

What I pass to `LinearRegression` or `LogisticRegression` as features:

$$\begin{array}{c} \\ 1 \\ 3 \\ 0 \\ 2 \\ 5 \\ 1 \\ 8 \end{array} \begin{array}{c} \begin{array}{ccc} z_1=-1 & z_2=3 & z_3=7 \quad z_4=11 \\ \texttt{[-1,3)} & \texttt{[3,7)} & \texttt{[7,11)} \end{array} \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{array}$$

Basis functions

Pick a collection of functions b_i and learn the model:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

My data:

$$\begin{pmatrix} 1 \\ 3 \\ 0 \\ 2 \\ 5 \\ 1 \\ 8 \end{pmatrix}$$

What I pass to `LinearRegression` or `LogisticRegression` as features:

$$\begin{pmatrix} b_1(1) & b_2(1) & \cdots & b_K(1) \\ b_1(3) & b_2(3) & \cdots & b_K(3) \\ b_1(0) & b_2(0) & \cdots & b_K(0) \\ b_1(2) & b_2(2) & \cdots & b_K(2) \\ b_1(5) & b_2(5) & \cdots & b_K(5) \\ b_1(1) & b_2(1) & \cdots & b_K(1) \\ b_1(8) & b_2(8) & \cdots & b_K(8) \end{pmatrix}$$

Section 2

Decision Trees

Subset of Hitters data

	Hits (x_1)	Years (x_2)	Salary ($\cancel{x}$)	LogSalary (y)
1	81	14	475.0	6.163315
2	130	3	480.0	6.173786
3	141	11	500.0	6.214608
4	87	2	91.5	4.516339
5	169	11	750.0	6.620073
...	...	...	...	...
317	127	5	700.0	6.551080
318	136	12	875.0	6.774224
319	126	6	385.0	5.953243
320	144	8	960.0	6.866933
321	170	11	1000.0	6.907755

predicted value = $\hat{y}$

actual salary predicted is $e^{\hat{y}}$

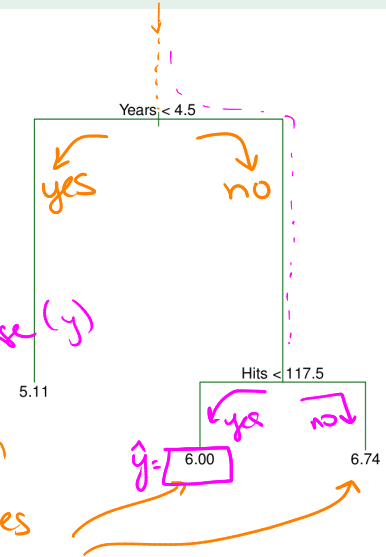
First decision tree example

	Hits	Years	LogSalary
1	81	14	6.163315
2	130	3	6.173786
3	141	11	6.214608
4	87	2	4.516339
5	169	11	6.620073
...	...	...	...
317	127	5	6.551080
318	136	12	6.774224
319	126	6	5.953243
320	144	8	6.866933
321	170	11	6.907755

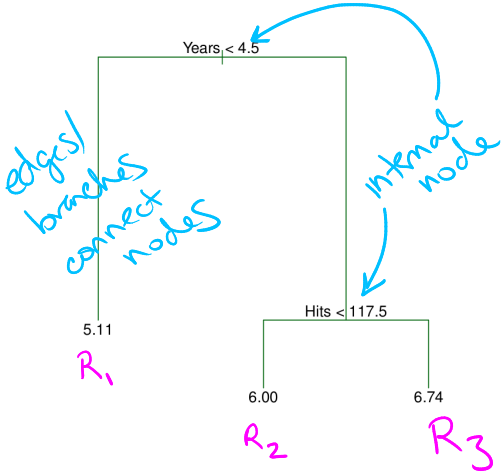
$\hat{y}$	$e^{\hat{y}}$
6.00	\$402,834
5.11	\$165,174

Only have 3 possible predictions

Prediction is mean response (y) of players w/ those properties



Regions defined by the tree

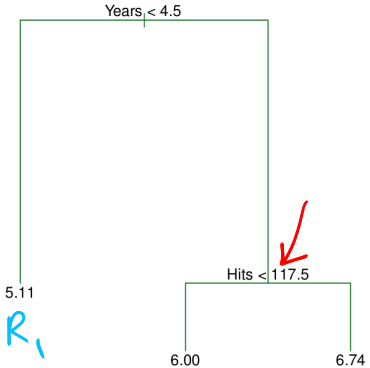


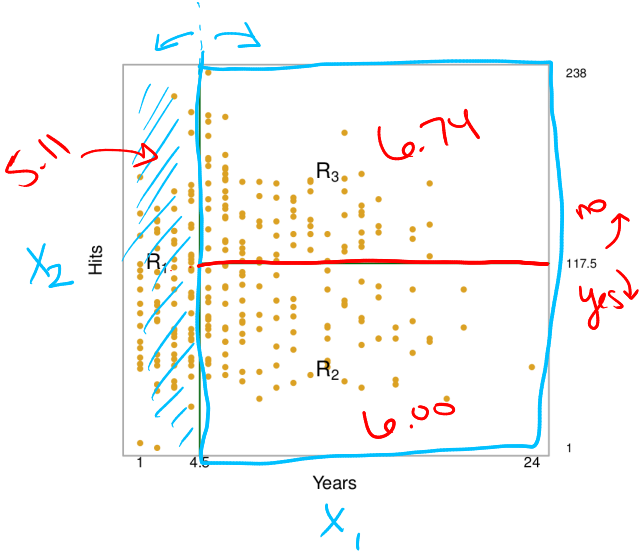
$R_1 = \{X \mid \text{years} < 4.5\}$

$R_2 = \{X \mid \text{years} \geq 4.5, \text{Hits} < 117.5\}$

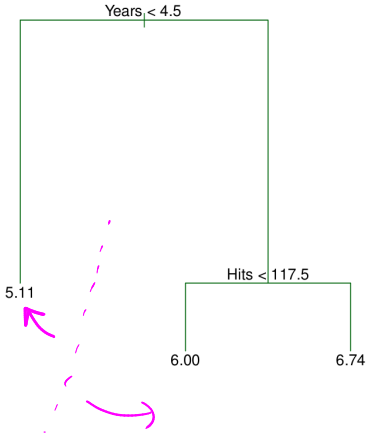
$R_3 = \{X \mid \text{years} \geq 4.5, \text{Hits} \geq 117.5\}$

Viewing Regions Defined by Tree





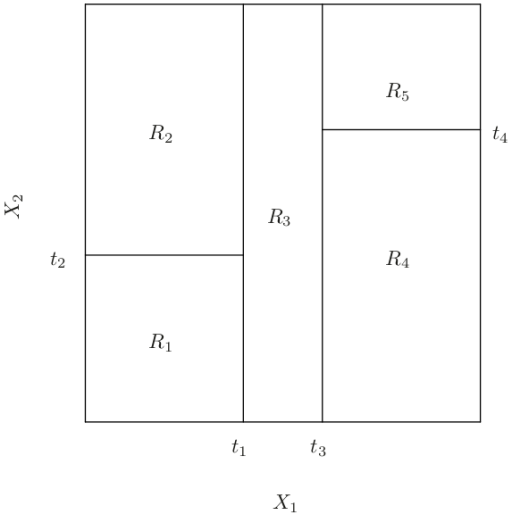
Interpretation of example



- Years most important factor for determing salary
- Players w/ less experience make less than more experience
- More experienced players, # of hits affects salary

How do we actually get the tree? Two steps

1. We divide the predictor space – that is, the set of possible values for $X_1, X_2, \cdots, X_p$ — into J distinct and non-overlapping regions, $R_1, R_2, \cdots, R_J$.
2. For every observation that falls into the region R_j, we make the same prediction = the mean of the response values for the training observations in R_j.



Step 1: How do we decide on R_js?

Goal:

Find boxes $R_1, \cdots, R_J$ that minimize

RSS

$$\sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

x_i in R_j

$\hat{y}_{R_j}$ = mean response for training observations in jth box

- Overfitting !
- Too many options to check
- Recursive binary splitting

top down

greedy algorithm

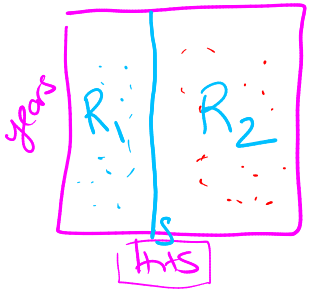
Recursive Binary Splitting

- Pick X_j ~ this
- Pick s so that splitting into $\{X \mid X_j < s\}$ and $\{X \mid X_j \geq s\}$ results in largest possible reduction in RSS

$$R_1(j,s) = \{X \mid X_j < s\}$$
$$R_2(j,s) = \{X \mid X_j \geq s\}$$

mean of y_i $x_i \in R_1(j,s)$

$$\underbrace{\sum_{i|x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2}_{\text{RSS}} + \sum_{i|x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2$$



Rinse and repeat

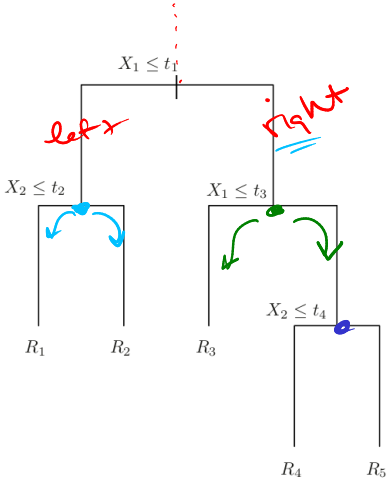
Pick j, Pick s

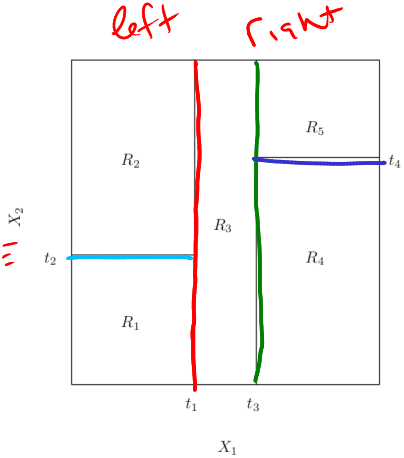
find the j, s

that makes RSS

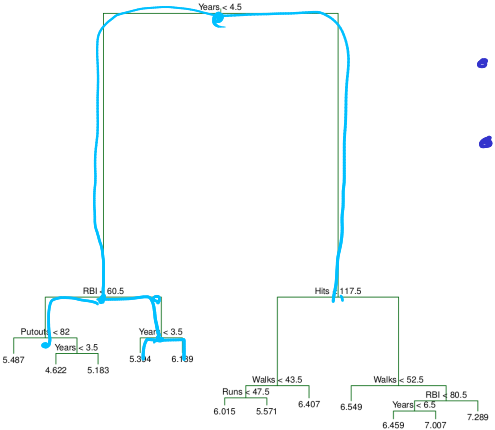
as small as possible

fix that + repeat





Pruning



- Big trees ⇒ overfitting
- Stop splitting earlier, meh
- Grow a big tree, prune it back.

↳ find subtree with best test error rate

Too many to test all subtrees

Weakest Link Pruning

Also called Cost complexity pruning

For every α, there is a subtree T that minimizes:

$$\underbrace{\sum_{m=1}^{|T|} \sum_{i|x_i \in R_m} (y_i - \hat{y}_{R_m})^2}_{} + \alpha |T|$$

- $|T|$ = number of terminal nodes of T, leaves
- R_m is rectangle for mth terminal node
- $\hat{y}_{R_m}$ is mean of training observations in R_m

- $\alpha = 0$ => returns whole tree
- Not obvious: Branches pruned from tree in nested & predictable fashion as $\alpha \to$ bigger

Algorithm version

Algorithm 8.1 *Building a Regression Tree*

1. Use recursive binary splitting to grow a large tree on the training data, stopping only when each terminal node has fewer than some minimum number of observations.

2. Apply cost complexity pruning to the large tree in order to obtain a sequence of best subtrees, as a function of α.

3. Use K-fold cross-validation to choose α. That is, divide the training observations into K folds. For each $k = 1, \ldots, K$:

   (a) Repeat Steps 1 and 2 on all but the kth fold of the training data.

   (b) Evaluate the mean squared prediction error on the data in the left-out kth fold, as a function of α.

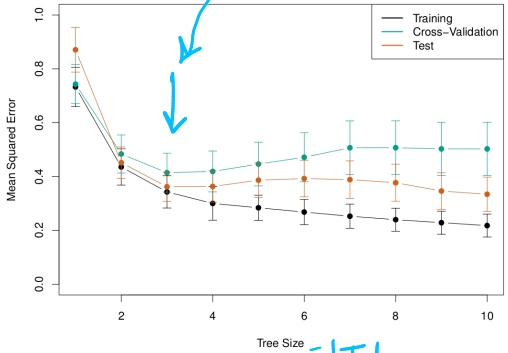
   Average the results for each value of α, and pick α to minimize the average error.

4. Return the subtree from Step 2 that corresponds to the chosen value of α.

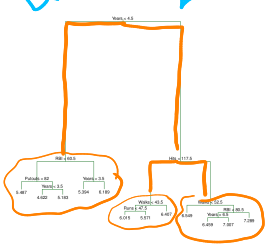
Messing with α

3 leaf tree

Ba tree



$= |T|$



Bagger

α ← increasing

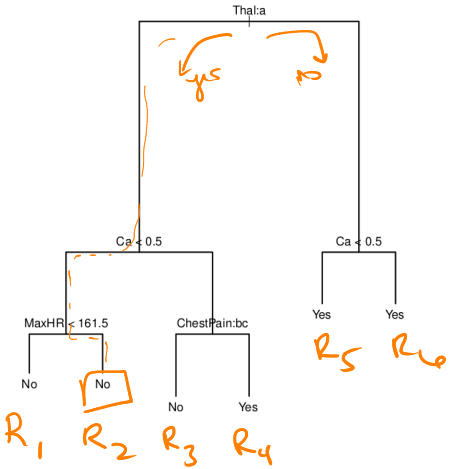
$\alpha = 0$

Section 3

Classification Decision Tree

Basic idea

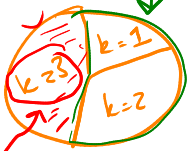
Heart data



Can't use RSS

Plan A: Use error rate

- $\hat{p}_{mk}$ = proportion of training observations in R_m from the kth class
- $E = 1 - \max_k(\hat{p}_{mk})$



$\hat{y}_m = k = 3$

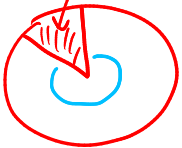
Very not sensitive

bad for growing trees

Gini index

$$G_m = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk})$$

Region R_m



- Measure of variance across all classes
- Small value if all $\hat{p}_{mk}$ are close to 0 or 1
- node mostly has obs. from one class

Entropy

$$D_m = \underbrace{-\sum_{k=1}^{K} \hat{p}_{mk} \log \hat{p}_{mk}}$$

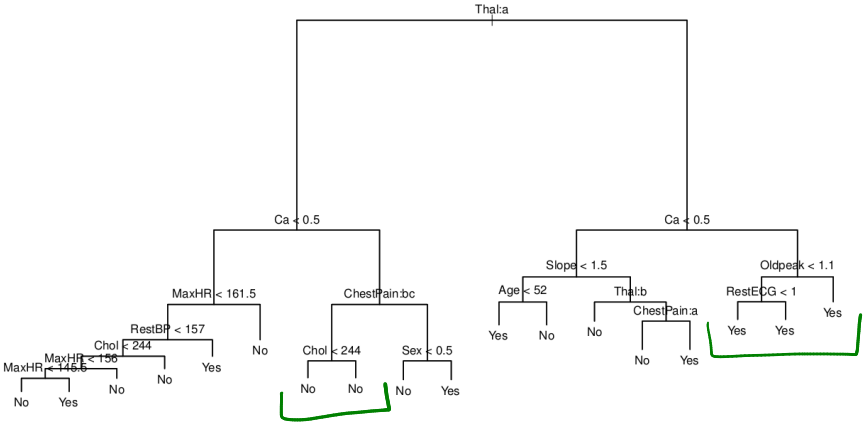
negative b/c

$0 \leq \hat{p}_{mk} \leq 1$

means D is positive

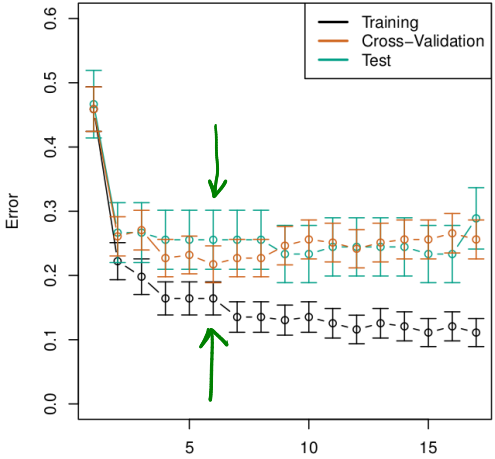
- Near zero if $\hat{p}_{mk}$ all near 0 or 1
- Small value if m^{th} node has majority one class

Example



↳ happens b/c improves entropy / Gini index

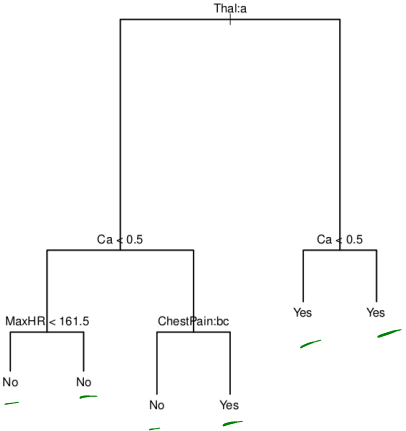
Pruning the example



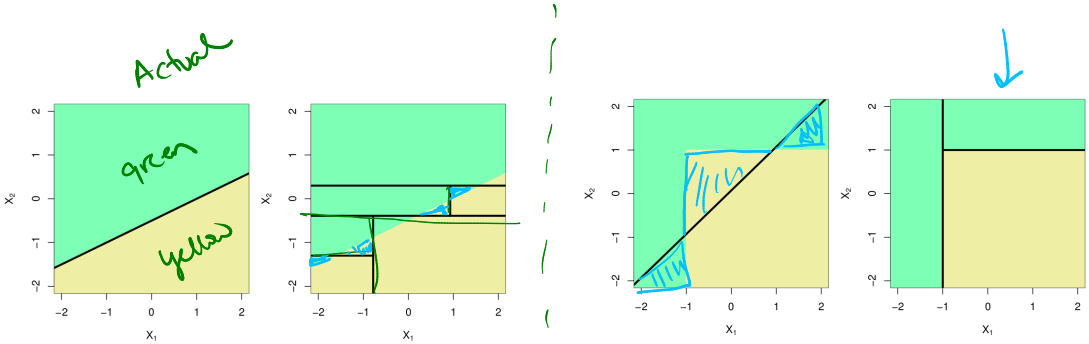
α = bigger

$= |T| = \#$ leaves

$\alpha = 0$



Linear models vs trees



tree model can't predict well

linear can't predict well

Pros/Cons

Pros:

- Easier to explain
- Graphical representation easy to interpret
- Easy to handle qualitative predictors

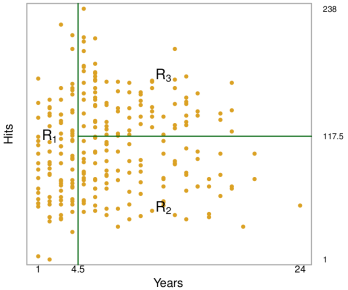
Cons:

- Not as accurate as ~~linear~~ other reg/classification
- Not robust
  - ↳ small change in data can cause large change in estimation

Fix: Aggregated trees

TL:DR

- Split into regions by greedily decreasing RSS
- Prune tree by using cost complexity
- Not robust - Next time, figure out how to aggregate trees



Next time

Next time:

- 8.2 Bagging, Random Forests, Boosting

This week:

- Mon 3/28: 8.1
- Weds 3/30: 8.2
- Fri 4/1: Lab: 8.3
  - HW #8 Due Friday

Overall, good length.
I added some examples that I think went well
Check the yes/no direction on the trees before this one